

Automation & QA Developer — Submission Brief

Candidate Submission · 25–26 May 2026 · App under test: demo.realworld.io (Conduit / RealWorld)

Task 1 — Web App QA & Debug Report

Tested the publicly hosted **Conduit** demo (React + Redux frontend, Node/Express backend). Exercised all main flows — sign-up, login, article CRUD, commenting, tagging, profile editing, pagination, and logout — plus edge cases (special characters, very long inputs, expired tokens, network throttling).

#	Issue	Severity
1	Invalid email accepted on sign-up — no client-side validation, server error swallowed	Critical
2	Post-login redirect always goes to global feed, losing original destination	High
3	Stale/corrupt JWT fails silently — no 401 interception, no auto-logout	High
4	XSS via unsanitised Markdown — dangerouslySetInnerHTML without DOMPurify	Critical
5	Offset-based pagination — no first/last controls, slow beyond page 10	Medium
6	Deleted-article tags persist indefinitely in sidebar	Low
7	No account deletion — GDPR data-right gap	Medium

Root-cause focus (Bug #4 — XSS): The Markdown renderer (marked.js) passes raw HTML into dangerouslySetInnerHTML without a sanitisation step. Fix: wrap with DOMPurify.sanitize() — one line, eliminates a Critical stored-XSS vector.

Task 2 — n8n Workflow: HackerNews Morning Brief

<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Node' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Node', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'What it does' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='What it does', textColor=Color(0,0,0,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>
Schedule Trigger	Fires every hour
HTTP – HN Top Stories	GET /v0/topstories.json → 500 ranked IDs (1st API call)
Code – Slice Top 5	Slices to first 5 IDs; emits one item per ID
HTTP – HN Story Detail	GET /v0/item/{id}.json for each ID (2nd API call — enrichment)
Code – Aggregate & Sort	Filters type=story, sorts by score desc, bundles into one output
IF – Score ≥ 100?	TRUE → Hot Digest path (🔥); FALSE → Quiet Digest path (🔇)
2× Code + Discord nodes	Format embed + POST to Discord webhook (one per branch)
Error path	All HTTP nodes use onError:continueErrorOutput → Format Error → Discord alert

APIs: HackerNews Firebase API (free, no auth key). Output: Discord webhook embed. No secrets hard-coded — all credentials stored in n8n Credentials store.

Bonus — Uptime Monitor

Pings **demo.realworld.io** every 5 minutes. If the first attempt returns non-200 or times out, two automatic retries fire before an alert is raised — avoiding false positives from transient blips. Downtime alerts go to the same Discord webhook as Task 2. Response time is tracked per check; a daily summary stub is included (wire to Google Sheets for production).

Deliverables

Task1_QA_Report.pdf	Bug table (7 issues) + root-cause analysis
Task2_Workflow.json	Importable n8n workflow (11 nodes)
Task2_Workflow_Screenshots.pdf	Canvas diagram + sample execution output
README_Task2.md	API rationale, transformation logic, error-path docs
Bonus_UptimeMonitor.json	Uptime monitor with 3-attempt retry
README.md	Top-level repo overview + setup guide
Submission_Brief.pdf	This document